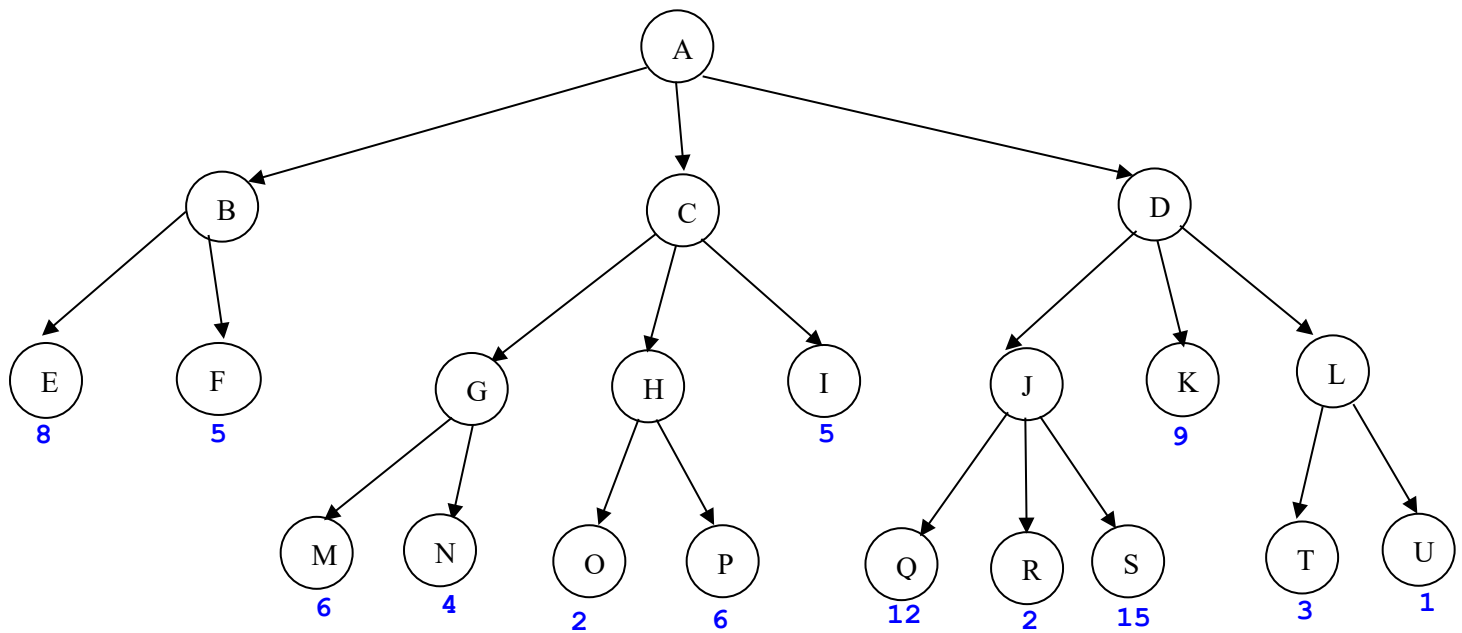

Problem 1 – Game Playing and Expected Values

Consider the following game tree in which the evaluation function values are shown below each leaf node. Assume that the root node corresponds to the minimising player. That is, the first player (MIN) is trying to minimise the final score. Assume that the search always visits children left-to-right.

1. Clearly indicate the MAX and MIN layers of the tree **and** use MINIMAX to determine the best first move for MIN.



2. In which order will the nodes be examined by the alpha-beta procedure? Which nodes will be pruned by the alpha-beta procedure?

Problem 2 – Propositional Logic

Your friend Tracy argues: “If Bob works hard then Bob has a lot of money. If Bob has a lot of money then Bob can buy the new iPhone. Bob cannot buy the new iPhone. Thus, Bob does not work hard.”

Regardless of whether the premises and conclusion are true, by using propositional logic, can you determine whether your friend Tracy’s argument is valid?

Problem 3 – Propositional Logic

About my friend Bob, we know the following (the agent knowledge base KB):

Bob is a student or a worker or both.

If Bob is a worker then Bob is at work.

Bob cannot be both at home and at work.

Bob is at home.

You can use the following vocabulary:

BiS = “*Bob is a student*”

BiW = “*Bob is a worker*”

W = “*Bob is at work*”

H = “*Bob is at home*”

1. Represent the knowledge base KB in propositional logic.
2. Using propositional logic and truth table, please answer True/False to the following questions and for each answer, list the models that support your answer):

- a. $KB \models W$
- b. $KB \models BiS$
- c. $KB \models BiW$
- d. $KB \models W \vee \neg BiW$

Problem 4 – Propositional Logic

Decide whether each of the following sentences is valid, unsatisfiable, or neither. Verify your decisions using truth tables.

- a. $(P \Rightarrow Q) \vee (Q \vee \neg P)$
- b. $((((A \wedge B) \Rightarrow C) \wedge (A \Rightarrow B)) \wedge A) \Rightarrow C$
- c. $((P \Rightarrow Q) \wedge P) \wedge \neg Q$
- d. $((P \Leftrightarrow Q) \wedge P) \vee \neg Q$

Problem 5 – AI Planning

Our agent performs planning to move cargoes between cities. The agent can load a cargo at a city into an empty cargo space in an airplane that is at the same city. The agent can also unload a cargo from an airplane. The agent can fly an airplane from a city to another city. The airplanes *Plane1* and *Plane2* are at *Melbourne* and the airplane *Plane3* is at *Sydney*. *Plane1* has two cargo spaces *CS11* and *CS12*. *Plane2* has three cargo spaces *CS21*, *CS22* and *CS23*. *Plane3* has two cargo spaces *CS31* and *CS32*. Cargo *C1* is currently occupying cargo space *CS12* in *Plane1*. Cargo spaces *CS11*, *CS21*, *CS22*, *CS23*, *CS31* and *CS32* are currently empty. Cargoes *C2* and *C3* are currently at *Melbourne*. Cargoes *C4* and *C5* are currently at *Sydney*.

The goal is to get the cargoes *C1*, *C2*, and *C3* to *Sydney* and to get the cargoes *C4* and *C5* to *Melbourne*.

1. Write down the initial state description and the agent's goals.
2. Write down STRIPS-style definitions of the three actions.
3. Write down a consistent partial-order plan (POP) with no open preconditions for this problem.

Problem 6 - Uncertain reasoning

Kangaroo Electronics is an electronics manufacturer that uses an AI system FPD to detect faulty products. The FPD system classifies a product into one of two bags: **Good** and **Bad**. When a faulty product is examined by FPD, it is classified as **Bad** by FPD with a probability of 0.98.

When a non-faulty product is examined by FPD, it is classified as **Bad** by FPD with a probability of 0.01. Statistics from Kangaroo Electronics shows that, on average, there is 1 in 200 products is faulty.

1. What is the probability that the next product is classified as **Bad** by FPD?
2. What is the probability that the next product is both faulty and classified as **Bad** by FPD?
3. What is the probability that the next product is non-faulty and classified as **Bad** by FPD?
4. A product is classified as **Bad** by FPD, what is the probability that it is actually faulty?